

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S6 (R,S) / S4 (PT) (R,S) Examination May 2024 (2019 Scheme)

Course Code: EET306**Course Name: POWER ELECTRONICS**

Max. Marks: 100

Duration: 3 Hours

Graph sheet may be provided

PART A*Answer all questions, each carries 3 marks.*

Marks

- | | | |
|----|--|-----|
| 1 | Sketch the static VI characteristics of SCR and define latching current and holding current. | (3) |
| 2 | Mention the advantages of wide band-gap power devices | (3) |
| 3 | Draw the circuit & output voltage waveform of a single half wave controlled rectifier. | (3) |
| 4 | Compare single phase full bridge converter and single phase semi converter | (3) |
| 5 | Explain the working of a single phase full bridge voltage source inverter with pure R load. Draw the output voltage waveform. | (3) |
| 6 | Write a short note on THD. | (3) |
| 7 | In a step up chopper the dc input voltage is of 100V. The MOSFET switch is having a switching frequency of 1kHz. Find the duty cycle and average dc output voltage if the turn on period of switch is 0.2ms. | (3) |
| 8 | Draw the waveform of inductor voltage of a buck dc-dc converter and obtain an expression for output dc voltage in terms of input voltage and duty cycle | (3) |
| 9 | Explain with relevant curves components of frictional load torque | (3) |
| 10 | Explain regenerative braking control in drives | (3) |

PART B*Answer one full question from each module, each carries 14 marks.***Module I**

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|----|--|-----|
| 11 | a) With the help of a diagram, explain the Two Transistor analogy of an SCR, with necessary equations. | (8) |
| | b) Explain the structural features of power MOSFET with a neat sketch | (6) |

OR

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|----|--|-----|
| 12 | a) Explain the mechanism of turning-on an SCR using its Turn-on characteristics. | (8) |
| | b) Explain how di/dt and dv/dt protection is accomplished in SCR. | (6) |

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Module II

- 13 a) Illustrate how a Thyristor based 1-phase fully controlled rectifier can be used to convert ac into variable dc. Draw the waveforms of output voltage, voltage across the thyristors & output current for both R and RL load at $\alpha=30^\circ$ (8)
- b) Obtain an expression for average dc output voltage of a 1-phase fully controlled rectifier for R load with firing angle α . For an sinusoidal AC input of 220 V rms at 50 Hz and a 20 ohm load resistor and delay angle is 40° Determine the average current in the load, (6)

OR

- 14 a) With the help of circuit diagram and relevant waveforms, explain the working of three phase fully controlled bridge converter, feeding RL load, with firing angle 60° . (10)
- b) What is the use of free-wheeling diode in the single-phase half wave converter feeding RL load? Explain the change in voltage waveform due to free-wheeling diode. (4)

Module III

- 15 a) Describe the operation of single phase AC voltage controller for R load with waveforms and derive expression for output rms voltage. (6)
- b) Illustrate the generation of sine pulse width modulated control signals for a single phase VSI with output voltage waveform. (8)

OR

- 16 Explain the operation of a 3-phase voltage source inverter with 180° mode of operation. Draw the phase voltages and line voltages across a star connected R load. (14)

Module IV

- 17 Describe the working of 4 quadrant choppers with proper circuit diagram and relevant waveforms (14)

OR

- 18 With circuit diagram and waveforms, describe the operation of a buck-boost dc-dc converter. Derive expressions for output dc voltage and the design equations for filter inductor & capacitor. (14)

Module V

- 19 a) What are the advantages of electric drives (6)
- b) Explain the working of a single phase full converter drive (8)

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OR

- 20 a) Explain the parts of electric drive with its block diagram (7)
- b) Why the stator voltage control for Induction motor drive has low efficiency? (7)
- Explain with relevant equations. Also draw the circuit diagram
